

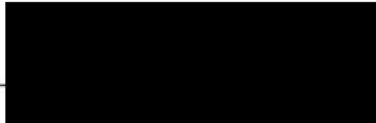
ChE422 Biochemical Engineering

2020 Midterm Exam

3/10/2020

2:30 PM - 3:45 PM

Name: \_\_\_\_\_



Instructions: This is a close-book and close-note exam.

52/100

**MULTIPLE CHOICE QUESTIONS (Part I):**

**Note: Only ONE answer is correct. Circle the correct answer. (2 pts for each problem)**

1. The bond between a phosphate group and the ribose sugar group in RNA is called which of the following?  
☒ (A) Phosphodiester linkage  
☐ (B) Glycosidic bond  
☐ (C) Peptide bond  
☐ (D) Amide bond
2. Which of the following plays a role in DNA transcription?  
☐ (A) DNA polymerase  
☒ (B) RNA polymerase  
☐ (C) DNAase I  
☐ (D) Guanyl transferase
3. What signals the end of transcription?  
☒ (A) Stop codon  
☒ (B) Terminator  
☐ (C) The end of the DNA chain  
☐ (D) RNA polymerase runs out
4. The start codon is:  
☐ (A) AUU      ☐ (B) UAA      ☒ (C) AUG      ☐ (D) UAG
5. Which of the following is not a component of a nucleotide?  
☐ (A) Phosphate group  
☒ (B) Anti-codon  
☐ (C) Ribose sugar  
☐ (D) Nitrogen base
6. What provides the energy that drives the addition of nucleotides to a growing DNA chain during replication?  
☒ (A) The hydrolysis of peptides  
☐ (B) The hydrolysis of the diphosphate group in dNTPs  
☐ (C) The hydrolysis of UTP  
☐ (D) The hydrolysis of GTP

7. Which of the following is the sequence of the -10 region of prokaryotic promoters?

- ~~(A) TATAAT~~ (B) TTGACA (C) TATA (D) CAA

8. How many base pairs make up a codon?

- (A) 1 (B) 2 (C) 3 (D) 4

9. How many amino acids are there?

- (A) 20 (B) 30 (C) 40 (D) 50

10. How many hydrogen bonds are formed between one A:T base pair?

- ~~(A) 1~~ (B) 2 (C) 3 (D) 4

### MULTIPLE SELECTION QUESTIONS (Part II):

**Note: ONE or MORE answers can be correct. Circle all correct answers. (3 pt for each problem)**

11. Which of the following are the protein secondary structures?

- (A) Hairpin (B)  $\alpha$ -helix (C)  $\beta$ -Sheet (D) Peptide bond

12. Which of the following in a plasmid controls the copy number of this plasmid in the host?

- (A) Antibiotic resistance gene  
(B) Multiple cloning sites  
(C) LacZ gene  
(D) Origin of replication (Ori)

13. Which of the followings are required for Polymerase chain reaction (PCR)?

- (A) DNA template (B) T4 ligase (C) DNA polymerase (Taq) (D) DNA primers

14. Which of the following methods are commonly used to transform plasmid into *E. coli* cells?

- (A) Cell fusion  
(B) Heat shock method  
(C) Electroporation  
(D) DNA injection by a needle

15. Which of the followings are the key components in a general Operon?

- (A) Promoter (B) Operator (C) phage gene (D) Structural genes

16. Which of the following methods are used in random mutagenesis?

- ☒ (A) Error prone PCR  
☒ (B) Chemical Deamination  
☐ (C) DNA shuffling  
☒ (D) *E. coli* XL1 Red method

17. Agarose gel electrophoresis is able to separate linear DNA fragments with which of the following parameter?

- ☒ (A) Charge ☒ (B) Size ☐ (C) Secondary structure ☐ (D) Conformation

18. In order to use pET expression system for protein expression, the host cell should provide with which of the following component for transcription?

- ☒ (A) Taq DNA polymerase ☒ (B) T7 polymerase ☐ (C) protease ☒ (D) Histag

19. Which of the following methods can be used to introduce the mutation at the specific sites in a gene?

- ☒ (A) UV irradiation  
☐ (B) Cassette mutagenesis  
☐ (C) PCR site-directed mutagenesis  
☒ (D) RACHITT method

20. Which of the following compound is used in pBAD expression system to induce protein expression?

- ☐ (A) Glucose ☒ (B) IPTG ☒ (C) L-arabinose ☐ (D) Lactose

Brief answer the following questions

1. What is Central Dogma of Molecular Biology? (5pts)

Central Dogma is like the central law of genetic code which is the basic law of translation process.

2. List three key differences between prokaryote and eukaryote? (5pts)

have or do not have membrane  
have or do not have ribosome  
have or do not have nuclei

-3  
the location of ~~protein~~  
DNA synthesizing is  
different.

3. Describe the general key components of an expression plasmid? (5pts)

we need the protein and DNA polymerase, RNA polymerase, helicase,  
peptidyl transferase.

-5

4. Describe the key differences between blunt-end ligation and sticky end ligation for cloning? (5pts)

Sticky end ligation for cloning needs ~~to~~ sticky end.  
blunt-end ligation doesn't need sticky end.

-5

5. List at least two methods to confirm the insertion of a gene into the expression plasmid? (5pts)

if we want to transfer the genetic information into cell, we can let cell carry the gene and let gene pass the membrane. ~~let cell carry the gene and let gene pass membrane.~~  
~~we can use PCR or NGS.~~

-5

6. Briefly describe blue-white screen? (10pts)

if we use it to determine ecoli  
the thing with blue color have e. coli  
the thing with white color don't have e. coli  
Also the water is the solution

-5

7. After you find an interesting gene in a new organism, briefly describe the procedures to let *E. coli* express this protein. (15pts)

First, we need to create a protein library. } a protein library can be created by in vitro transcription followed by translation of amplified DNA fragment which isolated by organism, that is, protein synthesis.

Protein synthesis is a process which organism synthesizes protein according to the genetic information from mRNA which transcribed of DNA.

Transcription is a process which the genetic information flow from DNA to RNA, that is, the process of RNA synthesis under the action of RNA polymerase, using a certain chain of double-stranded DNA as a template (template chain for transcription, coding chain not for transcription), using A, U, C, G as raw materials.

Translation is a process which decoding the nucleotide sequences of mature mRNA which transcribed of DNA, producing the corresponding gene sequences according to the central law of genetic code.

When we find out the gene sequences, we can use gene sequences to get the protein sequences. After we get protein sequences, we can let *E. coli* express this protein.